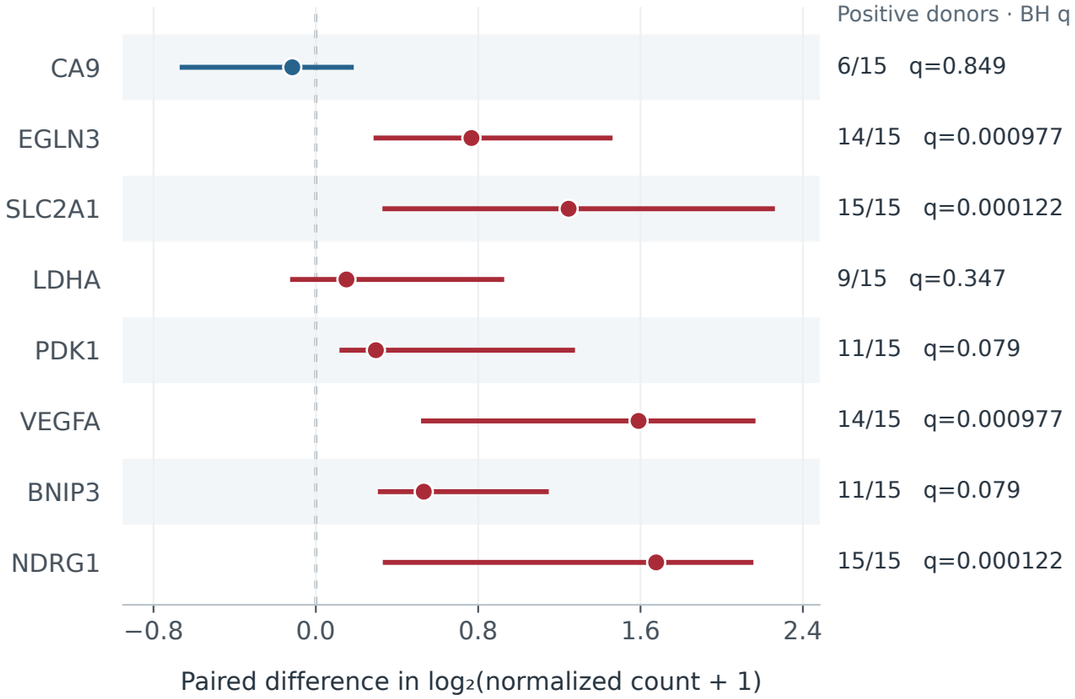
4 of 8 markers support the expected hypoxia direction across donors

GSE144424 · 15 human donors · iPSC-derived cardiomyocytes · exploratory external validation

Primary selection: one complete A–D experiment per donor; repeated differentiations do not add donors.

A · Hypoxia versus matched reference

Median donor difference and unadjusted 95% bootstrap interval



B · Six hours of hypoxia

1% oxygen for 6 h − matched 10% reference

Donor	CA9	EGLN3	SLC2A1	LDHA	PDK1	VEGFA	BNIP3	NDRG1
18499	-0.7	+0.3	+0.3	-0.2	-0.3	+0.2	-0.0	+0.0
18505	-0.3	+0.1	+0.3	-0.1	-0.2	+0.5	-0.0	+0.6
18511	-0.1	+1.5	+1.7	+0.9	+1.3	+1.6	+1.1	+2.1
18520	+1.1	+2.1	+3.2	+1.2	+1.5	+2.2	+1.9	+3.2
18852	-1.1	+1.2	+1.8	+0.2	+0.2	+2.2	+0.5	+1.7
18855	+0.6	+1.8	+2.6	+1.0	+1.3	+3.0	+1.8	+2.7
18858	+0.2	-0.1	+0.2	-0.0	+0.1	-0.3	-0.1	+0.3
18870	-0.8	+0.3	+0.3	-0.1	-0.1	+0.4	+0.4	+0.0
18912	-1.1	+0.5	+2.2	+0.0	+0.3	+1.8	+0.3	+1.7
19098	-0.1	+1.0	+0.4	+0.3	+0.3	+0.5	+0.5	+0.5
19101	+0.6	+2.2	+2.4	+0.7	+1.0	+2.8	+1.9	+3.1
19108	+0.4	+0.3	+1.2	-0.0	-0.6	+1.6	-0.0	+1.9
19116	-0.6	+0.3	+0.8	+0.3	+0.4	+0.8	+0.6	+0.1
19128	-0.1	+0.8	+1.0	-0.2	+1.3	+1.6	+0.9	+1.6
19160	+0.2	+2.3	+2.7	+1.2	+1.3	+2.7	+1.8	+3.1

C · Six hours after reoxygenation

6 h at 1%, then 6 h at 10% − earlier reference

Donor	CA9	EGLN3	SLC2A1	LDHA	PDK1	VEGFA	BNIP3	NDRG1
18499	-1.1	-1.3	-1.2	-0.6	-0.9	-0.8	-0.9	-1.2
18505	+0.1	-0.7	-0.2	-0.5	-0.7	-0.6	-0.6	+0.0
18511	-0.5	-0.3	+0.2	+0.5	-0.2	-0.5	-0.1	+0.7
18520	+0.9	-1.1	-0.3	+0.8	+0.7	+0.6	+1.1	+0.7
18852	-0.1	+0.2	+0.9	+0.5	+0.2	+0.7	+0.2	+0.9
18855	+1.1	+0.4	+1.1	+1.2	+0.7	+1.5	+1.6	+1.3
18858	-0.2	-1.7	-1.4	-0.7	-1.0	-1.1	-1.1	-1.3
18870	-1.2	-1.9	-0.4	-0.8	-1.5	-1.7	-1.1	-1.0
18912	+0.2	-0.4	+1.1	+0.3	-0.1	-0.3	+0.1	+0.7
19098	-0.1	-1.5	-1.0	-0.5	-0.8	-0.5	-0.7	-0.5
19101	+1.2	-0.0	+1.0	+0.5	-0.1	+1.0	+0.7	+2.1
19108	+0.6	-0.6	+0.3	+0.0	-0.7	-0.1	-0.4	+1.1
19116	-0.6	-0.9	+0.2	+0.1	-0.3	-0.5	-0.3	-0.4
19128	+0.0	+0.1	+1.4	+0.4	+0.9	+0.3	+1.7	+1.2
19160	+0.5	+0.1	+0.9	+1.2	+0.6	+0.8	+1.3	+1.7

D · Twenty-four hours after reoxygenation

6 h at 1%, then 24 h at 10% − earlier reference

Donor	CA9	EGLN3	SLC2A1	LDHA	PDK1	VEGFA	BNIP3	NDRG1
18499	-1.0	-0.7	-1.5	-0.3	-0.5	-1.5	-0.5	-1.7
18505	-0.8	-1.2	-1.2	-1.2	-1.6	-1.4	-1.4	-1.4
18511	-0.1	-0.5	-1.0	-0.4	-0.6	-1.0	-1.2	-0.9
18520	+0.9	-0.5	-0.4	-0.2	-0.3	+0.2	-0.2	+0.0
18852	+0.2	-0.4	-0.7	-0.0	-0.3	+0.1	-0.4	-0.7
18855	+0.4	+0.1	+0.1	+0.6	+0.2	+0.9	+0.8	+0.1
18858	-0.1	-1.7	-1.9	-0.8	-1.0	-1.7	-1.3	-1.9
18870	-3.3	-2.5	-2.7	-1.7	-2.4	-2.0	-2.6	-3.0
18912	-1.1	-0.8	-0.5	+0.1	-0.4	-0.8	-0.5	-0.9
19098	-1.6	-2.1	-2.4	-1.4	-2.5	-1.0	-1.7	-2.1
19101	-0.2	-0.1	-0.1	-0.5	-0.7	+0.1	-0.8	-0.7
19108	+0.0	-0.5	-0.7	-0.3	-1.1	-0.6	-1.0	+0.0
19116	-0.6	-0.9	-1.1	-0.7	-0.8	-1.2	-1.0	-1.8
19128	+1.5	+0.1	+0.2	+0.2	+0.7	+0.6	+0.8	+0.2
19160	-0.8	+0.1	+0.2	+0.0	-0.3	+0.4	+0.0	+0.3

Heatmaps use the same donor order and symmetric scale: red = higher, blue = lower. Labels are rounded; signed zeros can represent small nonzero differences.

Panel A: one-sided exact sign tests; Benjamini–Hochberg q across eight planned markers. Intervals: 20,000 donor resamples, conditional on normalization.

Reoxygenation is compared with the earlier baseline; later time-matched reference controls are absent. Expression proximity does not establish functional recovery.

Median-of-ratios normalization within this dataset only. These differences are not on the Tsao TPM scale. Frozen DermTrace datasets, models and benchmark are unchanged.